

Industrial Statistics

- **Annual Survey of Industries**

- Industrial Statistics (IS) Wing of DQAD has the overall responsibility for planning, designing, processing and publishing data collected through Annual Survey of Industries (ASI), and also analytical work relating to it.
- is the principal source of industrial statistics of registered manufacturing sector in India
- covers only organized manufacturing sector comprising activities relating to manufacturing processes, repair services, generation, transmission, and distribution of electricity, gas, water supply and cold storage.
- covers all factories registered under sections 2m(i) and 2m(ii) of Factories Act, 1948 and also covers Bidi and Cigar manufacturing establishments.
- conducted annually** under the Collection of Statistics (COS) Act, 2008 (as amended in 2017) and the rules framed there-under in 2011
- For selection of units in the Annual Survey of Industries, **the lists of factories maintained by the Chief Inspectors of Factories are used as the sampling frame.**
- Used in NAS for GDP calculations; Useful input for SDP; Helpful to monitor growth of org. mfg. sector

Def.:

ASI Schedule: ASI schedule is the basic tool to collect required data from the units selected for the survey. The schedule for ASI, at present, has two parts. Part-I of ASI schedule, processed at the IS Wing, DQAD Kolkata, aims to collect data on assets and liabilities, employment and labour cost, receipts, expenses, input items: indigenous and imported, products and by-products manufactured, distributive expenses, etc. Part-II of ASI schedule is processed by the Labour Bureau. It aims to collect data on different aspects of labour statistics, namely, working days, man-days worked, absenteeism, labour turnover, man-hours worked etc.

Reference Period: The reference period (year) for ASI is the financial year (April-March) for all items of Part – I & II. For example, reference period for ASI 2019-20 is the financial year 1 st April 2019 – 31st March 2020.

Survey Period: The actual survey period for ASI is generally from the month of November to July in the subsequent year following the reference period.

For data collection in ASI, an establishment approach (and not enterprise approach) is followed. An institutional unit in its capacity as a producer of goods and services is known as an enterprise. An enterprise may be engaged in one or more economic activities at one or more locations. An enterprise may be a sole legal unit. On the other hand, an establishment is defined as an enterprise or part of an enterprise that is situated in a single location and in which only a single productive activity is carried out or in which the principal productive activity accounts for most of the value of products and by-products.

The primary unit of enumeration in the survey is a factory in the case of manufacturing industries, a workshop in the case of repair services, an undertaking or a licensee in the case of electricity, gas and water supply undertakings and an establishment in the case of bidi and cigar industries.

ASI is a statutory survey and so the units/entrepreneurs are to submit the returns along with the balance sheet and other relevant documents within the prescribed period after the receipt of notice failing which penalty process may be initiated in accordance with the COS Act.

Sampling Design: According to the new sampling design, all the units in the updated frame are divided into two parts – Central Sample and State Sample. The Central Sample consists of two schemes: Census and Sample. Under Census scheme, all the units are surveyed.

(1) Census Scheme

(2) Sample Scheme: All the remaining units in the frame are considered under Sample Scheme. For all the states, each stratum is formed on the basis of State x District x Sector x 3-digit NIC2008. The units are arranged in descending order of their total number of employees. Samples are drawn using Circular Systematic Sampling technique for this scheme. An even number of units with a minimum of 4 units are selected and distributed in four subsamples. It may be noted that in certain cases each of 4 sub-samples from a particular stratum may not have equal number of units.

(3) Out of these 4 sub-samples, two pre-assigned sub-samples (1 & 3) are given to NSO (FOD) and the other two-subsamples (2 & 4) are given to concerned State/UT for data collection.

(4) All census units plus all the units belonging to the two sub-samples given to NSO (FOD) are treated as the Central Sample.

(5) All census units plus all the units belonging to the two sub-samples given to State/UT are treated as the State Sample. Hence, State/UT has to use Census Units (collected by NSO (FOD) and processed by IS Wing, DQAD) along with their subsamples while deriving the state level estimates for their respective State/UT based on State Sample.

(6) All census units plus all the units belonging to the two sub-samples given to NSO (FOD) plus all the units belonging to the two sub-samples given to State/UT are required for obtaining pooled estimates based on Central Sample and State Sample with increased sample size.

- **Economic Census**

-by ESD

-is the complete count of all non-farm economic establishments located within the geographical boundary of India

-provides an updated sampling frame for follow-up enterprise surveys undertaken for detailed and comprehensive analysis of different economic sectors in the country.

-Main Purpose:

- Generate updated frame of establishments for follow-up surveys
- To ascertain the structure, activity and performance of different sectors of the economy
- Preparation of frame for development of National Business Register

-In view of the rapid changes that occur in the unorganized sector of non-agricultural economy, the economic census is conducted periodically in order to update the information from time to time

-The Follow-up Enterprise Surveys (FuS) of the EC are periodically carried out by the MoS&PI to collect detailed information about employment, fixed assets, working capital, receipts, expenditure, gross value added, etc. from the enterprises.

-1st: 1977 ; 2nd: 1980 ; 3rd: 1990 ; 4th: 1998 ; 5th: 2005 ; 6th: 2013 ; 7th: 2019-20

-EC covers all establishments irrespective of their size of employments
-covers all active units both in formal and informal sectors
-scheme on Economic Census (EC) and Follow-up Enterprise Surveys was launched by the CSO in 1976 with two main objectives namely,
(a) To provide a frame (list) from which samples could be drawn for collecting detailed information; (b) To provide at regular intervals some basic information on location, type of activity, nature of operation, etc. about the enterprises in the country and number of persons employed by them at the level of villages, and census enumeration blocks, i.e. EBs in case of large villages and urban areas (except in the recent EC in which Urban Frame Survey blocks of NSSO were used).

- ASI vs Follow-up Enterprise Survey of Economic Census:

(a) Data on registered manufacturing and repairing units are collected through the Annual Survey of Industries and on unregistered manufacturing and repairing units, through the Followup Surveys of the Economic Census.
(b) For selection of units in the Annual Survey of Industries, the lists of factories maintained by the Chief Inspectors of Factories are used as the sampling frame. On the other hand, the Follow-up Enterprise Surveys (FuS), which adopt a stratified two or three-stage sampling design, with villages and urban blocks as the first-stage units (FSUs) and unregistered manufacturing and repairing units as the ultimate stage sampling units, generally use the list of villages and blocks with information on number of enterprises and workers as per the Economic Census as the sampling frame for selection of FSUs.
(c) Collection of data from the units in the Annual Survey of Industries is record-based. But in the Follow-up Enterprise Surveys, data are generally collected by interviewing the respondents, as most units do not maintain any books of accounts.

- **Index of Industrial Production(IIP)**

- ☐ Base: 2011-12
- ☐ released by ESD,NSO
- ☐ Quick estimates released on 12th of every month (or previous working day if 20th is a holiday) with a lag of 6 weeks.
- ☐ to measure changes over time in the volume of output of industrial sector and it should not reflect any changes in the price.
- ☐ Laspeyre's formula is used for compiling IIP.
- ☐ In case of all India IIP, value figures of output are deflated by WPI.
- ☐ based on secondary data, wherein data is supplied to ESD(NSO) from 14 source agencies in various dept./ministries
- ☐ covers 3-sectors: (Sector-based classification)

Sector	Weightage	No. of items	Source of Data
Manufacturing	77.633%	405	(DPIIT) Dept for Promotion of Industry and Internal trade
Mining	14.373%	1	Indian Bureau of Mines(IBM), Nagpur
Electricity	7.994%	1	Central Electricity Authority

- Use-based Classifications:

Primary Goods(34%)	Intermediate Goods(17%)	Consumer Non-durables(15%)
Consumer Durables(13%)	Infra/Construction Goods(12%)	Capital Goods(8%)

- ☐ 8 core industries viz. Refinery Products, Electricity, Steel, Coal, Crude Oil, Natural Gas, Cement, Fertilizers constitute 40.27% weight in overall IIP

- **Index of Eight Core Industries**

-base: 2011

-released by OoEA,

-to measure changes over time in the volume of eight core sectors

-covers 8 sectors

-comprises 40.27% weight in IIP

Sector	Weight (in %)
Petroleum Refinery	28.04
Electricity Generation	19.85
Steel	17.92
Coal	10.33
Natural gas	8.98
Crude Oil Petroleum	6.88
Cement	5.73
Fertilizers	2.63

- What are the reasons for divergence in the growth rates of ASI and IIP?

Since the Annual Survey of Industries (ASI) is the main source of long term industrial statistics, and IIP is a monthly indicator based on items and factories selected from ASI, it is obvious that the growth rates of manufacturing sector based on the two datasets will be compared. However, while comparing the growth rates, the following must be borne in mind:

✓IIP is based on a fixed set of items and factories chosen in the base period whereas ASI is a record-based survey of establishments registered under Factories Act, 1948 in which the sampling frame and the sampled establishments undergo significant changes. As a result, ASI captures information of new items and factories whereas IIP does not.

✓IIP is based on a much smaller sample of factories as compared to that of ASI.

✓Growth rates in IIP are based on volume of production whereas growth rates in ASI are derived on the basis of Value Added (Output – Input). As these parameters are conceptually different, the resultant growth rates are also different.

✓Establishments selected in IIP are generally larger in size whereas ASI establishments cover both large and smaller units. As a result, the growth rates in IIP are comparatively lower as the smaller units have a small base and hence higher growth.